

Brandon LiWang

MECHANICAL ENGINEERING

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CORE COMPETENCIES

- CAD (SolidWorks, Fusion)
- Composites manufacturing (fiberglass, carbon fiber, etc)
- **Software:** MATLAB, Python, C++, Simulink, Linux, Git
- 3D Printing Optimization
- Rapid Prototyping & Fabrication
- Woodshop (Table saw, dremel, etc)
- **Hardware:** Soldering, Microcontroller (Arduino, Raspberry Pi)
- Sensor & Embedded Integration
- Data Processing & Analysis
- System Architecture & Documentation
- Machine Learning and Neural Networks
- Drone Piloting (107 Certified)

WORK EXPERIENCE

UNIVERSITY OF CALIFORNIA, MERCED – MESA LAB, JUNE 2024 TO AUGUST 2025

ENGINEERING RESEARCH INTERN

- Built an autonomous, methane-seeking UGV
- Designed modular sensor housings and mounting architecture to allow rapid payload swaps and dependable outdoor operation;
- Used CAD + rapid prototyping to build and refine functional assemblies under tight timelines;
- Implemented a two-phase waypoint algorithm (area-enclosing sweep → concentration-guided convergence) for methane source-seeking;
- Integrated mechanical, electrical, and software subsystems (Raspberry Pi, Pixhawk, sensors) into a cohesive autonomous platform;
- Contributed in a **fast-paced, multi-role lab environment**, shifting between and finishing several projects

Machine Learning High-Five Counter:

- Implemented a machine-learning pipeline using a neural network to recognize and count high-fives
- Collected motion data via an Arduino-mounted accelerometer and transformed time-series signals into ML features
- Trained and deployed a real-time inference system using transfer learning for high-five classification

EDUCATION

UNDERGRADUATE OF MECHANICAL ENGINEERING

University of California, Berkeley

PUBLICATIONS

- *A low-cost autonomous diffusing gas source detection system with a UGV and empirical observability Gramian* - IEEE - [Published]

COLLABORATION & LEADERSHIP

- Worked with undergraduate, graduate, and post-doctoral researchers
- Trained undergraduates on various introductory projects and technical skills
- Learned valuable skills such as:
- **Working under pressure**
- Leading project tasks
- Working under the lead of those more experienced
- **Taking constructive criticism**

ADDITIONAL INFORMATION

Citizenship: U.S. Citizen | Open to relocation & travel

Languages: English, Spanish

Interests: Autonomous sensing platforms, spacecraft mechanisms, high-reliability hardware, fishing, running, and music